

Rupak Ganvir

+91 91465 09525 | rupakganvir12@gmail.com | LinkedIn | GitHub | Portfolio

Education

Indian Institute of Technology (IIT) Kanpur

2025 – Present

M.Tech. in Geoinformatics

- Coursework: Spatial Data Science, Probability & Statistics, Numerical Methods

Priyadarshini College of Engineering, Nagpur

2019 – 2023

B.Tech. in Computer Science & Engineering

- Coursework: Data Structures & Algorithms, OOP, DBMS, Computer Networks

Projects

Fraud Detection System – End-to-End ML Pipeline + FastAPI | *Python, XGBoost, SMOTE, Scikit-learn*

- Built a fraud detection pipeline on the IEEE-CIS dataset (590K+ transactions), implementing Logistic Regression from scratch in NumPy and improving AUC-ROC from 0.784 to 0.904 using XGBoost.
- Addressed severe class imbalance with SMOTE, engineered transaction and email-domain features, and deployed a FastAPI service for fraud scoring, risk categorization, and Folium-based geospatial visualization.

Fake Medicine Detection – Decentralized Blockchain System | *Ethereum, Solidity, React, Web3.js, Smart Contract*

- Built a full-stack Ethereum-based drug verification system (Solidity + React + Web3.js + MySQL) for tamper-proof pharmaceutical supply chain tracking, recognized by faculty as a high-impact, production-grade capstone.

Publications

Pharmacy Fraud Detection using Blockchain by Scanning QR Code – IJARIIIE, Vol. 9, Issue 3, ISSN: 2395-4396

- Co-designed a decentralized Ethereum system using smart contracts and QR-code scanning to detect counterfeit pharmaceuticals, with a full-stack implementation spanning Solidity, React, Web3.js, Android, and MySQL.

Technical Skills

AI/ML: NumPy, Pandas, Matplotlib, XGBoost, Scikit-learn, SMOTE, Logistic Regression, FastAPI

Languages: Python, C++, JavaScript, SQL

Frontend/Backend: React, Node.js, MongoDB, REST APIs

Tools: Git, Linux, Jupyter, Kaggle, Google Colab

Experience

Software Engineer (Frontend & Automation)

Apr 2024 – Oct 2024

ASP OL Media Pvt. Ltd.

Nagpur, India

- Built React/HTML/CSS/JS frontends for a multi-campaign email automation platform serving millions of users, achieving zero cross-client rendering issues across Gmail, Outlook & mobile clients.
- Engineered an SMTP delivery pipeline with queue management and bounce analytics, cutting delivery failures by 20% through proactive server health monitoring.
- Owned the full design-to-deployment lifecycle of production dashboards under tight daily deadlines while maintaining consistent uptime.

Frontend Developer Intern

Apr 2023 – Sep 2023

DevKalp Technologies

Remote

- Delivered a hospital management dashboard (React + Tailwind) for 500+ patients with real-time doctor availability and booking, shipped to production within a 6-month contract.
- Built responsive UI for 2 e-commerce clients as part of a 4-member agile team using Git branching and component-level code reviews.

Achievements & Certifications

- **Published Research** – Peer-reviewed co-author, IJARIIIE (June 2023) on blockchain-based pharmaceutical fraud detection.
- **Competitive Programming** – CodeChef 3-Star; solved 150+ LeetCode problems with a medium/hard focus.
- **Udemy C++ Certification** – C++ fundamentals, OOP, and STL.